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Skewness and Concentration in Nordic Equity Markets: Implications for Portfolio Construction

MSc. in Apllied Financial Mathematics

Leo Jozic
William Öbrink

Supervisor : Taras Bodnar
Examiner : Jörgen Blomvall

LINKÖPING UNIVERSITY

Linköpings universitet
SE–581 83 Linköping
+46 13 28 10 00 , www.liu.se

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Nomenclature

Symbol	Description
$R_{i,t}$	Arithmetic return of asset i at time t
$r_{i,t}$	Logarithmic (continuously compounded) return of asset i at time t
$P_{i,t}$	Price of asset i at time t
$R_{p,t}$	Portfolio return at time t
$R_i^{lifetime}$	Lifetime return for asset i
w_i	Weight of asset i in a portfolio
N	Number of assets in a portfolio
$\bar{R}$	Mean return
σ	Standard deviation (volatility)
SK	Skewness of the return distribution
x_i	Individual observation
$\bar{x}$	Sample mean
s	Sample standard deviation
RI_t	Total return index at time t
DY_t	Dividend yield at time t
$WC_{i,t}$	Wealth creation of firm i at time t
WC_i	Total wealth creation of firm i
WC_{market}	Total wealth creation of the market
$M_{i,t}$	Market capitalization of firm i at time t
$R_{f,t}$	Risk-free rate at time t
β_i	Beta (systematic risk) of asset i
σ_i	Volatility of asset i
σ_m	Volatility of the market
$\sigma_{i,m}$	Covariance between asset i and the market
μ_i	Expected return of asset i
μ_m	Expected market return
$\nu_{i,t}$	Drift term in the return process for asset i at time t
$\xi_{i,t}$	Stochastic shock (random component) for asset i at time t
ϵ_i	Standardized random variable for asset i
Σ	Covariance matrix of asset returns
σ_p	Portfolio volatility

Symbol	**Description**
$RC_{i,t}$	Risk contribution of asset i at time t
d_t	Return difference between two portfolios
$\bar{d}$	Mean return difference
T	Number of observations
t	t-statistic
p	p-value
Var_{NW}	Newey–West variance estimator
γ_k	Autocovariance at lag k
L	Lag length in the Newey–West estimator
$V_{p,t}$	Total portfolio value at time t

1 Introduction

Research conducted in U.S. equity markets shows that the majority of buy-and-hold common stock returns are outperformed by returns on one-month U.S. Treasury bills held over the same investment horizon (Bessembinder, 2018). This finding challenges the conventional view of equity markets as broadly rewarding for long-term investors and suggests that aggregate market returns may obscure substantial heterogeneity in individual stock outcomes.

Equity markets play a central role in long-term wealth accumulation, pension systems, and capital allocation. Market performance is therefore often summarized using long-run average returns or broad market indices. While such measures provide a useful aggregate perspective, they offer limited insight into how returns are generated at the individual stock level. Aggregate market returns in the U.S. mask a highly uneven distribution of outcomes across firms, where a small number of stocks generate large positive returns while the majority deliver modest or negative performance over long horizons (Bessembinder, 2018).

Previous research has documented this pattern extensively for the U.S. equity market. In particular, Professor Hendrik Bessembinder shows in his research (Bessembinder, 2018) that most individual stocks underperform certain risk-free assets over their lifetimes, while nearly all long-term net wealth creation is attributable to a small fraction of firms. These results imply that individual stock returns are highly right-skewed and that aggregate market performance is driven by a limited number of extreme winners.

However, comparable evidence for Nordic equity markets is limited. This is notable given that Nordic markets differ from the U.S. in terms of market size, sector composition, ownership structures and institutional characteristics. It is therefore of interest to examine whether a similar return structure is present in Nordic equity markets.

Understanding the distribution of individual stock returns has important implications for portfolio construction and active asset management. In the presence of highly skewed return distributions, diversification primarily serves to increase the probability of capturing extreme positive outcomes rather than merely reducing volatility (Reinholtz, Fernbach, and Langhe, 2016). Similarly, the performance of concentrated portfolios and stock-picking strategies must be assessed in a setting where underperformance relative to the market is the most

likely outcome, despite the existence of a small number of highly successful investments.

By combining an empirical analysis of individual stock return distributions, a wealth-creation framework, and a portfolio-level evaluation, this study contributes to the literature by extending existing evidence beyond the U.S. market and by providing insights into the consequences of skewed return distributions for investors in Nordic equity markets.

1.1 Kvartil

This thesis is conducted in collaboration with Kvartil, an asset management firm founded in 2018 that combines data, statistical analysis, and systematic rules in its investment process to enhance decision-making and portfolio construction. Kvartil was founded by former employees of Brummer & Partners, bringing extensive experience from quantitative asset management. The team at Kvartil has contributed with domain expertise, valuable insights, and guidance throughout this thesis.

1.2 Purpose

The purpose of this study is to examine the achieved returns from portfolios constructed with different methods in a market environment characterized by positively skewed and concentrated individual stock returns.

1.3 Research questions

1. Do pooled return distributions in Nordic equity markets display significant positive skewness across different investment horizons?
2. How concentrated is long-term wealth creation across firms in the Nordic equity markets?
3. How do different portfolio construction methods affect portfolio returns in a market characterized by skewed and concentrated stock returns, and how do these portfolios compare in the sense of return to a broad Nordic market index and to randomly constructed portfolios?

1.4 Delimitations

To maintain a clear and well-defined scope, and to ensure consistency and availability of the data, this thesis applies several delimitations to the study design.

1.4.1 Nordic markets

This study is exclusively focused on the Nordic equity markets. By limiting the scope to the Nordic markets, the analysis aims to provide a more internally consistent setting in which observed return patterns can be attributed to underlying market dynamics rather than cross-market structural differences. The following markets are the ones considered in this study:

- Sweden
  - Nasdaq Stockholm Stock Exchange
  - Spotlight Stock Market
- Denmark

  - Nasdaq Copenhagen Stock Exchange
- Norway
  - Euronext Oslo Stock Exchange
- Finland
  - Nasdaq Helsinki Stock Exchange
- International
  - Nordic Growth Market

Since the study also aims to provide a measure on how different portfolio returns compare to a broader Nordic market index return, it is of importance to choose an index with respect to the Nordic market. The following market index has been selected for this study:

- Nordic index
  - STOXX Nordic

which is a subset of the STOXX Europe 600. STOXX Nordic index represents large, mid, and small capitalization companies in Sweden, Norway, Denmark, and Finland (STOXX Ltd., 2026).

1.4.2 Time horizon and return periods

The analysis considers one-year, five-year, and lifetime return horizons in order to capture both short- to long-term return dynamics and long-run wealth creation. For the one-year and five-year horizons, the sample is restricted to the most recent 35-year period. This choice reflects a trade-off between obtaining a sufficiently large number of observations and maintaining relevance to the current and recent market environment.

1.4.3 Data selection

For the one-year and five-year return analyses, the dataset includes return observations from periods during which companies were publicly listed and actively traded on the Nordic equity markets. In contrast, the lifetime return analysis includes all companies that have been listed on the Nordic markets at some point in the sample time, covering their entire listing history from initial listing to delisting or the end of the sample period.

2 Background

This chapter provides the background necessary to understand the theoretical aspects of stock returns and their implications for investment outcomes. It begins by introducing the concept of return distributions and the statistical characteristics commonly observed in financial data. Particular attention is given to skewness, which reflects the asymmetric nature of stock returns and highlights the presence of extreme outcomes. The chapter then examines how such asymmetries translate into concentration in long-term market performance, where a limited number of firms account for a disproportionate share of total wealth creation. The concept of wealth creation is introduced as a framework for quantifying the economic contribution of individual firms to aggregate market outcomes. Finally, the chapter discusses the implications of the empirically observed skewness in pooled return observations, together with the concentration of long-term wealth creation, for portfolio construction. This provides the conceptual foundation for the analysis that follows, which investigates how different portfolio construction approaches perform in an environment characterized by asymmetric and highly concentrated return outcomes.

2.1 Return distribution on financial markets

In financial markets, asset returns are commonly analyzed through their statistical distributions. A return distribution describes how frequently different return outcomes occur over a given horizon and provides a compact representation of both typical and extreme outcomes. Understanding the shape of return distributions is therefore a fundamental prerequisite for analyzing risk, performance, and investment outcomes.

In empirical studies, returns are typically measured as arithmetic or logarithmic price changes over fixed intervals, such as daily, monthly, or annual horizons (Miskolczi, 2017). The collection of these observations over time forms an empirical return distribution for a specific firm, which can be summarized using descriptive statistics such as the mean, median, skewness, and standard deviation. While these summary measures provide useful information, they do not fully characterize the distributional shape of returns.

In much of classical financial theory, asset prices are commonly modeled as following a geometric Brownian motion. Under this assumption, continuously compounded logarithmic

returns are normally distributed, while prices themselves follow a lognormal process (Mota, 2012). However, a large amount of empirical evidence suggests that observed logarithmic return distributions frequently deviate from normality. Observed return distributions often exhibit features such as asymmetry, heavy tails, and excess kurtosis, meaning that extreme outcomes occur more frequently than predicted by a normal distribution (Cont, 2001). These deviations are observed across asset classes, markets, and time periods, and become particularly pronounced when returns are measured over longer horizons or at the level of individual stocks.

As a result, empirical return distributions are typically analyzed using non-parametric tools, such as histograms, which allows one to visualize the data without imposing strong distributional assumptions. These tools provide a flexible way to visualize and summarize return behavior and serve as a natural starting point for further distributional analysis.

This study adopts such an empirical perspective when examining stock returns in Nordic equity markets. Rather than assuming a particular parametric form, the analysis begins by documenting the observed distributional properties of returns across different horizons. This approach allows for a transparent assessment of how closely empirical return distributions align with standard theoretical assumptions and provides the foundation for subsequent analysis of asymmetry, concentration, and portfolio-level implications.

2.2 Skewness in return distribution

Skewness is a fundamental property of return distributions that captures the degree of asymmetry around the central tendency. In a positively skewed distribution, the right tail is longer or heavier than the left tail, implying that a small number of observations take on exceptionally large positive values, while the majority of observations are clustered around lower or moderate outcomes. (Doane and Seward, 2011)

In the context of financial returns, when individual stocks are considered as separate observations, their long-term return outcomes exhibit a highly positively skewed cross-sectional distribution. Empirically, the distribution of individual U.S stock returns is characterized by many small losses or modest gains, combined with a small number of extreme winners (Bessembinder, 2018). As a result, the distribution of returns is asymmetric, and measures of central tendency such as the mean and the median can differ substantially. The distinction between mean and median outcomes is economically important. In a positively skewed distribution, the mean is pulled upward by rare but extreme positive outcomes, while the median reflects the outcome experienced by a typical observation (Doane and Seward, 2011). This feature implies that overall market performance may be driven by a limited number of exceptional firms rather than reflecting the typical outcome across stocks.

Recent empirical evidence highlights the extent to which individual stock returns exhibit such skewed behavior. Bessembinder (2018) documents that, over long horizons, the majority of U.S. stocks generate returns below those of one month Treasury bills, while a relatively small subset of firms accounts for nearly all positive net wealth creation in the market. These findings illustrate that aggregate market performance is driven by a small number of extreme winners, while the typical stock contributes little to long-term wealth creation.

Importantly, skewness is a defining feature of equity return distributions with direct economic implications. When returns are highly skewed, investment outcomes become sensitive to the inclusion or exclusion of a small number of extreme winners. This property challenges

intuitive interpretations based on average performance and motivates a closer examination of how skewed stock-level returns aggregate at the market and portfolio level.

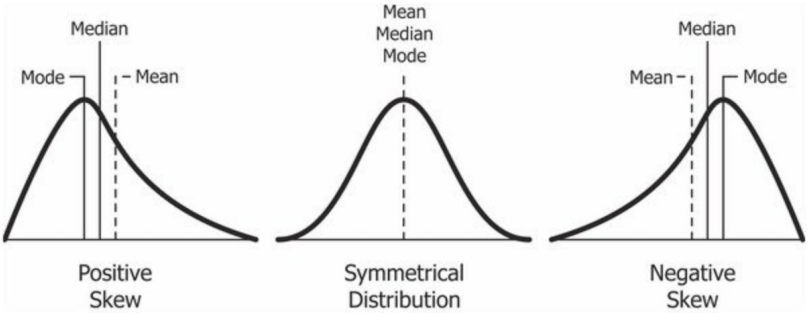

Figure 2.1: Illustration of different skewed distributions. Source: (Wikimedia Commons contributors, n.d.).

2.3 Long-term stock return and concentration

While skewness describes asymmetry in return distributions, it does not fully capture the economic importance of extreme outcomes. To assess their market-level relevance, it is necessary to examine how long-term stock returns and wealth creation are distributed across firms. When returns are measured over long horizons, such as several years or over a firm's entire publicly traded lifetime, each stock contributes a single cumulative outcome. Empirical evidence shows that this distribution is highly uneven, where most stocks generate modest or negative long-term returns, while a small number of firms achieve exceptionally large gains (Bessembinder, 2018).

This uneven contribution is captured by the concept of concentration, which refers to how total market wealth creation is distributed across firms. Aggregate market performance may therefore be driven by a limited subset of companies, even when the overall market exhibits strong long-term returns. Bessembinder (2018) formalizes this idea using the concept of wealth creation, defined as the cumulative value generated by a firm's equity in excess of a risk-free investment over its lifetime. Using this measure, he documents that nearly all net wealth creation in the U.S. equity market can be attributed to a small fraction of firms, while the typical stock underperforms risk-free assets (Bessembinder, 2018).

Applying the wealth creation framework further enables an investigation of the temporal structure of skewness. Because wealth creation accumulates multiplicatively over time, it offers a direct way to examine whether return concentration increases with the investment horizon. This allows the study to distinguish between skewness arising from the cross-sectional distribution of single-period returns and skewness that is amplified through long-term compounding effects.

Wealth creation thus provides a direct link between skewness at the individual stock level and concentration in aggregate market outcomes. This perspective motivates an examination of whether similar patterns of concentration are present in Nordic equity markets and sets the stage for analyzing their implications for portfolio-level outcomes.

2.4 Implications for portfolio management

The presence of skewness and concentration in the return distribution have an impact when it comes to portfolio construction. When the long-term returns are unevenly distributed across firms, aggregate market performance is driven by a relatively small number of extreme positive outcomes. As a result, investment outcomes depend critically on whether a portfolio includes the firms responsible for the majority of long-term value creation.

In such an environment, the interpretation of average returns becomes less informative for individual investors. While the mean return across stocks may be positive, due to the positive skewness in the return distribution, the typical stock may deliver substantially lower performance. This implies that investors holding only a limited number of securities face a significant probability of underperforming the broader market, even over long investment horizons. These characteristics highlight the role of diversification as a mechanism for increasing the likelihood of capturing the small subset of firms that generate the majority of long-term gains. Diversification, rather than solely reducing volatility (Reinholtz, Fernbach, and Langhe, 2016), improves the probability that a portfolio includes the extreme positive outcomes that drive aggregate market performance.

At the same time, different portfolio construction approaches may result in different exposure to these extreme outcomes. By sorting firms into size segments (small-, mid-, and large-cap) and varying the number of stocks included in each portfolio, the study examines how skewness, momentum, and diversification influence portfolio return distributions. This approach makes it possible to analyze how these factors interact and shape outcomes in a market characterized by concentrated wealth creation and skewed return distributions. This framework motivates an investigation into whether different portfolio construction rules lead to systematically different return distributions.

3 Methodology

This chapter outlines the methodological framework of the study. The objective is to examine whether Nordic equity markets exhibit skewed return distributions and concentrated wealth creation, and to analyze how these structural characteristics affect portfolio outcomes under different construction rules. The chapter begins with a review of relevant literature and a description of the data collection and processing procedures. It then presents the modeling framework, where skewness and wealth concentration are documented and portfolios are constructed based on predefined selection rules. Finally, an evaluation framework is introduced to assess differences between rule-based portfolios and to benchmark their performance against randomly formed portfolios. Together, these components provide a structured approach for linking stock-level return characteristics to portfolio-level performance in an environment characterized by asymmetric and concentrated return outcomes. All empirical analysis is conducted using data from Refinitiv Workspace and implemented in Python.

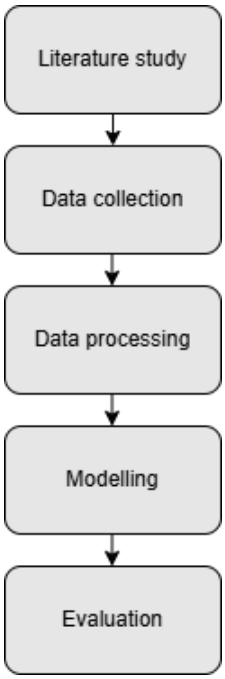

Figure 3.1: Overview of the methodology of the study.

3.1 Literature study

This study builds upon prior empirical research examining the distributional properties of individual stock returns and their implications for aggregate market outcomes. A contribution in this area is a study conducted by Bessembinder (2018), who analyzes the distribution of long-term stock returns using comprehensive data on U.S. equities. His research provides a framework for quantifying concentration in long-term equity returns and serves as a key reference point for this study. In addition to Bessembinders work, other research papers have been used that document that individual stock returns exhibit highly asymmetric distributions, characterized by positive skewness and the presence of extreme positive outcomes. This study builds on these approaches by applying similar distributional analyses to Nordic equity markets, specifically examining the empirical distribution of stock returns, quantify wealth creation and concentration, and construct rule-based portfolios to analyze how portfolio size and selection rules influence the distribution of portfolio outcomes.

3.2 Data collection

All necessary data for this study are obtained from Refinitiv Workspace.

3.2.1 Markets

To examine whether return skewness is present in Nordic equity markets, the analysis is based on firms listed on the major Nordic exchanges. Rather than focusing on a single country, the study covers multiple markets in order to capture the overall structure of return distributions in the Nordic region and to reduce the influence of country-specific effects.

The collected dataset includes all companies that were listed in any of the selected Nordic equity markets at some point during the sample period. This implies that the sample is defined by listing location rather than country of incorporation. Consequently, a subset of

the firms in the study may be registered or headquartered in another country while being publicly traded on a Nordic exchange.

From an investment perspective, what matters is where the security is traded and accessible to market participants, not necessarily where the firm is legally incorporated. By taking this approach the results describe the structure of returns in Nordic exchanges, rather than the performance of purely Nordic-originated firms. This distinction is important for interpretation, particularly when comparing findings to studies based on domestic firm samples in other regions.

The following markets are the ones included in the study:

- Sweden
  - Nasdaq Stockholm Stock Exchange
  - Spotlight Stock Market
- Denmark
  - Nasdaq Copenhagen Stock Exchange
- Norway
  - Euronext Oslo Stock Exchange
- Finland
  - Nasdaq Helsinki Stock Exchange
- International
  - Nordic Growth Market

3.2.2 Returns

This study is based on arithmetic returns generated by publicly traded companies listed on the Nordic exchanges described in Section 3.2.1. For firms that were delisted during the sample period, the reason for delisting is identified and treated according to the defined data processing procedure 3.3.

In finance, returns are commonly expressed as logarithmic (continuously compounded) returns. This convention stems from their analytical convenience, particularly their time-additive property and their compatibility with geometric averaging and continuous compounding (Campbell, Lo, and MacKinlay, 1997). Accordingly, and in line with established practice, this study also employs logarithmic returns in the analysis.

3.3 Data processing

To enable a consistent and reliable empirical analysis, the collected data were subject to a structured process of cleaning, processing, and formatting. The data preparation was conducted in Excel, including the use of built-in functions and VBA scripts to manage large datasets and automate repetitive procedures.

A central component of the data processing concerned the treatment of firms that were delisted during the sample period. For each delisted company, the reason for delisting was

identified in order to ensure an economically correct measurement of the final return. In cases of bankruptcy or liquidation, the terminal return was recorded as a total loss, reflecting the fact that shareholders effectively lost their invested capital. In contrast, when a firm was delisted due to an acquisition or merger, the final observed return was calculated based on the last available price, including any compensation received by shareholders. This approach is similarly applied in (Bessembinder, 2018).

This distinction is crucial, as improper treatment of delistings may lead to survivorship bias (Financial Edge Training, 2022) in the distribution of returns, particularly in studies examining skewness and the concentration of long-term wealth creation.

3.4 Modeling

The modeling framework links stock-level return characteristics to portfolio-level outcomes. Specifically, the analysis proceeds in two stages. First, skewness and wealth concentration in individual stock returns are documented in order to establish the structural properties of the Nordic equity market. Second, portfolios are constructed based on predefined selection rules to examine how these structural features translate into portfolio performance.

3.4.1 Observation of skewness and concentration among returns

The first step of this study is to conduct an empirical observation of the distributional properties of stock returns in the Nordic market and their implications for wealth creation. This empirical observation can be divided into two stages.

First, the distribution of stock return observations is examined across different investment horizons, including one-year, five-year, and lifetime returns. The analysis focuses on the empirical shape of these distributions using graphical methods such as histograms and QQ-plots. By documenting asymmetry and tail behavior without imposing parametric assumptions, this step establishes whether return outcomes exhibit pronounced skewness and uneven dispersion.

Second, firm-level wealth creation is evaluated to assess the concentration of long-term wealth creation within the Nordic market. Wealth creation is defined as the cumulative value generated by a firm's change in market capitalization in excess of a risk-free investment over its publicly traded lifetime. Firms are then compared based on their relative contribution to total market wealth creation, allowing for an assessment of how unevenly wealth creation is distributed across firms.

3.4.2 Portfolio construction

To examine the implications of skewness in individual stock returns, a range of portfolios is constructed and analyzed. The objective is to evaluate differences in performance across portfolios and to examine how their return distributions vary depending on underlying stock characteristics.

Portfolios are constructed along several key dimensions, including market capitalization segment, level of diversification, weighting scheme, and selection rule. Specifically, stocks are grouped according to market capitalization, such as small-, mid-, and large-cap segments, to account for potential size-related effects. Portfolio size is controlled by varying the number of constituent stocks, allowing for an analysis of diversification effects across different levels. In addition, different weighting schemes are applied, such as equal weighting and alternative

weighting approaches, to examine how the allocation of capital influences portfolio performance. Finally, portfolios are formed based on distinct selection rules, including rule-based strategies derived from statistical characteristics of returns, as well as randomly generated portfolios used as a benchmark.

By systematically combining these dimensions, a structured set of portfolios is obtained, enabling a comprehensive analysis of how portfolio construction choices affect return distributions and performance outcomes.

3.5 Evaluation

This section outlines the framework used to evaluate the performance of the constructed portfolios. The evaluation is conducted in two complementary stages. First, rule-based portfolios are compared statistically to assess whether systematic differences in construction rules translate into differences in average returns. Second, each rule-based portfolio is evaluated relative to a benchmark distribution of randomly formed portfolios in order to assess whether its performance is distinguishable from random selection. The rule-based portfolios are also compared relative to a market index and the risk-free rate.

3.5.1 Statistical comparison of rule-based portfolios

To compare portfolios constructed under different rules, daily return differences between two portfolios are computed over the evaluation period, which is a period of the most recent ten years. The mean return difference is then tested for statistical significance. The statistical significance of the estimated mean return differences depends critically on the relationship between the average return differential and its volatility. Since standard inference relies on the ratio between the mean and its standard error, which scales with the sample standard deviation and the square root of the number of observations, the amount of historical data required to detect statistically significant differences varies across portfolio pairs. In particular, when return differentials exhibit high volatility relative to their mean, longer time series are necessary to obtain sufficient statistical power.

Because daily financial returns may exhibit heteroskedasticity and serial dependence, inference is conducted using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey and West, 1987). This approach provides robust standard errors that remain valid under general forms of time-series dependence. Statistical significance is assessed using the corresponding t-statistic and p-values derived from the asymptotic distribution of the estimator. The sign of the estimated mean difference determines which portfolio exhibits higher average performance, while statistical significance indicates whether the observed difference is unlikely to be driven by sampling variation.

3.5.2 Benchmarking against randomly formed portfolios, market index and risk-free rate

In addition to pairwise statistical tests, each rule-based portfolio is evaluated relative to a benchmark distribution of randomly formed portfolios. For each portfolio configuration, defined by size segment, diversification level, skewness and momentum, a large number of portfolios are generated through random selection of stocks under identical constraints. Furthermore, comparisons against a broad Nordic market index is made to provide a concrete benchmark that reflects aggregate market performance. To evaluate returns generated from portfolios against risk-free investments, the returns will be compared towards a risk-free rate.

This Monte Carlo simulation approach produces an empirical distribution of portfolio

outcomes that reflects the range of performance attainable through random diversification within the same investment universe. The realized performance of the rule-based portfolio is then evaluated relative to this distribution, for example by assessing its percentile rank or by computing the proportion of randomly formed portfolios that achieve higher performance. The evaluation framework combines statistical inference between rule-based portfolios with simulation-based benchmarking against random portfolios. This enables a comprehensive assessment of whether systematic portfolio construction rules meaningfully affect investment outcomes in a market characterized by skewed and concentrated stock returns.

4 Theory

This chapter presents the theoretical concepts that provide the foundation for the empirical analysis conducted in this study. The purpose is to introduce the key financial and statistical concepts required to analyze the distributional properties of stock returns, the concentration of wealth creation across firms, and the evaluation of portfolio performance. The chapter begins by introducing common measures of asset returns, including arithmetic and logarithmic returns, which form the basis for empirical analysis of financial time series. These return measures are then considered in the context of return distributions, where particular attention is given to skewness as a measure of asymmetry. Understanding the distributional shape of stock returns is essential for analyzing the presence of extreme outcomes and the role they play in driving long-term investment results.

Building on the statistical properties of return distributions, the concept of wealth creation is introduced as a framework for measuring the economic contribution of individual firms to aggregate market outcomes. This framework allows the total value generated in the equity market to be decomposed into firm-level contributions and provides a way to analyze how concentrated long-term value creation is across companies.

Finally, the chapter presents the statistical tools used to evaluate portfolio performance in the empirical analysis. In particular, the paired t-test with Newey–West adjusted standard errors is introduced as a method for testing differences in average portfolio returns, and Monte Carlo simulation is presented as a framework for benchmarking rule-based portfolios against randomly formed portfolios. Together, these theoretical concepts provide the analytical foundation for the empirical investigation of skewness, wealth concentration, and portfolio outcomes in Nordic equity markets.

4.1 Arithmetic returns

Arithmetic returns represent one of the most commonly used measures of asset performance in financial analysis (Missiakoulis, Vasiliou, and Eriotis, 2008). The arithmetic return measures the simple percentage change in the price of an asset between two points in time and provides an intuitive measure of the gain or loss experienced by an investor over a given period.

Formally, the arithmetic return for asset i over period t is defined as

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}, \tag{4.1}$$

where $P_{i,t}$ denotes the asset price at time t and $P_{i,t-1}$ denotes the asset price in the previous period (Hudson and Gregoriou, 2010).

Arithmetic returns are widely used in empirical finance due to their straightforward interpretation and their compatibility with many statistical methods. In particular, arithmetic returns are additive across assets within a single time period, which makes them suitable for calculating portfolio returns. For a portfolio consisting of n assets with weights w_i, the portfolio return in period t can be written as

$$R_{p,t} = \sum_{i=1}^{n} w_i R_{i,t}. \tag{4.2}$$

This linear aggregation property makes arithmetic returns especially convenient in portfolio analysis and cross-sectional empirical studies.

4.2 Logarithmic returns

Logarithmic returns, often referred to as continuously compounded returns (Hudson and Gregoriou, 2010), represent an alternative way of measuring asset performance. Instead of measuring the simple percentage change in price, logarithmic returns measure the logarithmic change in asset prices between two points in time. This transformation provides several useful mathematical properties that make logarithmic returns widely used in financial modeling and econometrics.

The logarithmic return for asset i over period t is defined as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{4.3}$$

where $P_{i,t}$ denotes the asset price at time t and $P_{i,t-1}$ denotes the price in the previous period (Hudson and Gregoriou, 2010). The logarithmic return can also be expressed in terms of the arithmetic return $R_{i,t}$ as

$$r_{i,t} = \ln(1 + R_{i,t}). \tag{4.4}$$

One important advantage of logarithmic returns is that they are additive across time. For a sequence of returns over T periods, the cumulative logarithmic return is simply the sum of the individual period returns

$$r_{i,(1,T)} = \sum_{t=1}^{T} r_{i,t}. \tag{4.5}$$

This additive property simplifies the analysis of multi-period investment performance and therefore makes logarithmic returns particularly convenient in a financial setting (Hudson and Gregoriou, 2010).

4.3 Total return index

The Total Return Index (TRI) provided in Refinitiv Workspace represents the cumulative growth in the value of a common stock over a specified period (Refinitiv, 2024). The index assumes that all dividends are reinvested in the underlying security. Specifically, dividends are used to purchase additional shares at the closing price on the ex-dividend date, thereby capturing both capital appreciation and reinvested dividend income in the total return measure.

$$RI_t = RI_{t-1} \cdot \frac{PI_t}{PI_{t-1}} \cdot \left(1 + \frac{DY_t}{100} \cdot \frac{1}{N}\right) \tag{4.6}$$

where

RI_t = return index on day t

PI_t = price index on day t

DY_t = dividend yield % on day t

4.4 Descriptive statistics

To summarize the empirical return distributions, several descriptive statistics are used, including the mean, median, standard deviation, and skewness.

The mean return represents the average outcome and is defined as

$$\bar{R} = \frac{1}{N} \sum_{i=1}^{N} R_i, \tag{4.7}$$

where R_i denotes individual return observations and N is the number of observations.

The median return represents the central value of the ordered return distribution and is less sensitive to extreme observations than the mean.

The standard deviation measures the dispersion of returns around the mean and is defined as

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (R_i - \bar{R})^2}. \tag{4.8}$$

While the mean reflects the average return, the median provides the return experienced by a typical observation. In the presence of skewed return distributions, these measures may differ substantially, making both important for the analysis.

4.5 Skewness

Skewness is a statistical measure that describes the asymmetry of a probability distribution around its mean. It provides information about how observations are distributed in the tails relative to the center of the distribution. In a perfectly symmetrical distribution, such as the normal distribution, skewness equals zero and the mean, median, and mode coincide. When a distribution is skewed, the mass of observations is unevenly distributed around the mean, which causes these central tendency measures to differ. A positively skewed distribution is characterized by a longer right tail, indicating that a small number of large positive observations pull the mean above the median. Conversely, a negatively skewed distribution

has a longer left tail, where extreme negative observations pull the mean below the median. (OpenStax, 2019)

In empirical applications, skewness is estimated using the sample skewness measure according to

$$SK = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^3, \tag{4.9}$$

where n is the number of observations, $\bar{x}$ is the sample mean, and s is the sample standard deviation.

4.5.1 Histogram

A histogram is a graphical representation used to illustrate the distribution of a dataset. The range of observed values is divided into a number of intervals, referred to as bins, and the frequency of observations within each interval is displayed as bars. Histograms provide an intuitive way to visualize the shape of a distribution, including characteristics such as central tendency, dispersion, skewness, and the presence of extreme values. (Ruppert, 2011)

4.5.2 QQ-plots

A quantile–quantile plot (QQ-plot) is a graphical tool used to compare the distribution of a dataset with a theoretical distribution, or to compare two empirical distributions. The plot is constructed by pairing the quantiles of the sample data with the corresponding quantiles of the reference distribution. If the two distributions are similar, the points will lie approximately along a straight diagonal line. Deviations from this line indicate differences between the distributions, such as skewness, heavy tails, or other departures from the assumed distributional form. (Ruppert, 2011)

4.6 Momentum

The momentum effect is one of the most well-documented anomalies in financial markets. It refers to the empirical observation that assets which have performed well in the recent past tend to continue to outperform in the near future, while poorly performing assets tend to underperform. This phenomenon was first documented by Jegadeesh and Titman (1993), who show that strategies based on buying past winners and selling past losers generate statistically significant excess returns. In this study, momentum is based on performance of a company over a year. If a positive return is generated the company is considered having momentum. Companies with momentum are then ordered from highest to lowest yearly return.

4.7 Wealth creation

An important implication of the asymmetric distribution of individual stock returns is that long-term value creation in equity markets tends to be highly concentrated among a small subset of firms. While many firms generate modest or negative long-run returns, a limited number of companies account for a large share of the total wealth created in the market. To analyze this phenomenon, Bessembinder (2018) introduces a framework for measuring firm-level wealth creation relative to a risk-free investment.

The concept of wealth creation measures the monetary value generated by a firm's equity in excess of the return that could have been earned by investing the same capital in a risk-free asset. Instead of focusing solely on percentage returns, this approach evaluates the

dollar value added to investors over time. This distinction is important because firms differ substantially in size, and therefore a given percentage return may translate into very different levels of wealth creation depending on the firm's market capitalization.

Formally, the wealth created by firm i during period t can be expressed as

$$WC_{i,t} = M_{i,t-1}\left(R_{i,t} - R_{f,t}\right), \tag{4.10}$$

where $M_{i,t-1}$ denotes the market capitalization of firm i at the beginning of period t, $R_{i,t}$ represents the total return of the firm's equity during the period, and $R_{f,t}$ denotes the return on the risk-free asset. This formulation captures the excess value generated for investors during the period relative to the risk-free benchmark.

The total wealth created by a firm over its lifetime is obtained by aggregating these periodic contributions. In order to make contributions from different periods comparable, each period's wealth creation is typically expressed in terms of a common end-of-sample value by compounding with the risk-free rate. The cumulative wealth creation for firm i can therefore be written as

$$WC_i = \sum_{t=1}^{T} WC_{i,t} \prod_{s=t+1}^{T} (1 + R_{f,s}), \tag{4.11}$$

where T denotes the final observation period.

This framework highlights that total wealth creation depends not only on the returns generated by a firm, but also on the amount of capital invested in the firm over time. Large firms that generate moderate returns may therefore contribute more to total market wealth creation than smaller firms that experience extremely high percentage returns.

Aggregating wealth creation across firms allows for the measurement of total wealth generated in the equity market. The total market wealth creation can be expressed as

$$WC_{market} = \sum_{i=1}^{N} WC_i, \tag{4.12}$$

where N denotes the number of firms in the market and WC_i represents the cumulative wealth creation of firm i. This aggregation captures the total value created by the equity market in excess of the risk-free benchmark over the sample period.

By comparing each firm's wealth creation to the total market wealth creation, it is possible to evaluate the relative contribution of individual firms to aggregate market value creation. This provides a framework for analyzing the concentration of wealth creation across firms and identifying the extent to which a small subset of companies accounts for a disproportionate share of the total value generated in the market.

In the context of this study, the concept of wealth creation provides a framework for quantifying how value generated in Nordic equity markets is distributed across firms. By measuring firm-level contributions to aggregate market wealth, it becomes possible to evaluate the extent to which long-term value creation is concentrated among a small subset of companies.

4.8 Time aspect in the skewness of return distributions

It is of interest to study the underlying causes of skewness in common stock returns. In particular, it is important to distinguish whether the observed skewness arises from statisti-

cal properties of return data or whether it is driven by a small number of firms generating extreme returns. Understanding this distinction is central for evaluating the role of skewness in equity markets and for assessing how it may influence portfolio outcomes.

Let $S_{i,t}$ denote the price of asset i at time t, and $S_{m,t}$ the level of the market index. The arithmetic returns of individual stocks and the market index are defined as

$$r_{i,t} = \frac{S_{i,t}}{S_{i,t-1}} - 1 \tag{4.13}$$

$$r_{m,t} = \frac{S_{m,t}}{S_{m,t-1}} - 1. \tag{4.14}$$

The expected returns are estimated from historical observations. According to Blomvall (2026) the market index the point estimator is defined as

$$\bar{r}_m = \frac{1}{(t_T - t_0)\Delta t} \sum_{t=t_0+1}^{t_T} r_{m,t}, \tag{4.15}$$

while the expected return for asset i is estimated as

$$\bar{r}_i = \frac{1}{(t_T^i - t_0^i)\Delta t} \sum_{t=t_0^i+1}^{t_T^i} r_{i,t}. \tag{4.16}$$

The volatility of the market index and individual assets can then be estimated using unbiased estimators of the return variance (Blomvall, 2026). The market volatility estimate is given by

$$\bar{\sigma}_m = \sqrt{\frac{1}{(t_T - t_0 - 1)\Delta t} \sum_{t=t_0+1}^{t_T} (r_{m,t} - \bar{r}_m)^2 \Delta t}, \tag{4.17}$$

and the volatility of asset i is estimated as

$$\bar{\sigma}_i = \sqrt{\frac{1}{(t_T^i - t_0^i - 1)\Delta t} \sum_{t=t_0^i+1}^{t_T^i} (r_{i,t} - \bar{r}_i)^2 \Delta t}. \tag{4.18}$$

The covariance between asset i and the market return is given by

$$\bar{\sigma}_{i,m} = \frac{1}{(t_T^i - t_0^i - 1)\Delta t} \sum_{t=t_0^i+1}^{t_T^i} (r_{i,t} - \bar{r}_i \Delta t)(r_{m,t} - \bar{r}_m \Delta t), \tag{4.19}$$

which yields the beta coefficient

$$\bar{\beta}_i = \frac{\bar{\sigma}_{i,m}}{\bar{\sigma}_m^2}. \tag{4.20}$$

Under the Capital Asset Pricing Model (CAPM), the expected return of asset i can be expressed as

$$\mu_{i,t} = r_{f,t} + \beta_{i,t}(\mu_{m,t} - r_{f,t}), \tag{4.21}$$

where $r_{f,t}$ denotes the risk-free rate.

It is assumed that the equity volatility $\sigma_{i,t} = \bar{\sigma}_i$ which gives the expected logarithmic return,

$$\nu_{i,t} = \frac{\ln(1 + \mu_{i,t}\Delta t) - \frac{\sigma_{i,t}^2 \Delta t}{2}}{\Delta t} \tag{4.22}$$

To analyze the statistical properties of returns, logarithmic returns are modeled as a stochastic process (Blomvall, 2026).

$$\ln \frac{S_{i,t}}{S_{i,t-1}} = \nu_{i,t}\Delta t + \xi_{i,t}\sqrt{\Delta t}, \tag{4.23}$$

where $\nu_{i,t}$ represents the drift component and $\xi_{i,t}$ a random shock with mean zero. Aggregating returns over the interval $[t_0^i, t_T^i]$ yields

$$\ln \frac{S_{i,t_T^i}}{S_{i,t_0^i}} = \nu_i T_i + \xi_i \sqrt{T_i}, \tag{4.24}$$

where

$$T_i = (t_T^i - t_0^i)\Delta t, \tag{4.25}$$

$$\nu_i = \frac{1}{T_i} \sum_{t=t_0^i+1}^{t_T^i} \nu_{i,t}\Delta t, \tag{4.26}$$

and

$$\xi_i = \frac{1}{\sqrt{T_i}} \sum_{t=t_0^i+1}^{t_T^i} \xi_{i,t}\sqrt{\Delta t}. \tag{4.27}$$

According to Blomvall (2026), normalizing the stochastic component by the volatility yields the standardized variable

$$\epsilon_i = \frac{\xi_i}{\sigma_i}, \tag{4.28}$$

which under the model assumptions should follow a standard normal distribution

$$\epsilon_i \sim N(0, 1). \tag{4.29}$$

This transformation enables the use of standard statistical tools to assess whether equity returns follow a normal distribution and to evaluate the presence of skewness in return distributions.

4.9 Portfolio weighting schemes

The weighting scheme is a fundamental component of portfolio construction, as it determines how capital is allocated across assets and how risk and return are distributed within the portfolio. This study considers three commonly used weighting approaches: equal weighting, market capitalization weighting, and risk parity weighting.

4.9.1 Equally weighted portfolios

In an equally weighted portfolio, each asset is assigned the same weight, such that

$$w_i = \frac{1}{N}, \quad \forall i = 1, \ldots, N, \tag{4.30}$$

where N denotes the number of assets in the portfolio.

This approach is described by Bodie, Kane, and Marcus (2014) in the book Investments. It is stated that the weighting scheme provides a simple and transparent benchmark and ensures maximum diversification in terms of capital allocation. However, since assets typically exhibit different levels of volatility, equal weighting does not imply equal risk contributions. As a result, more volatile assets tend to have a larger impact on overall portfolio risk.

4.9.2 Market capitalization weighted portfolios

In a market capitalization weighted portfolio, each asset is weighted proportionally to its market value. Let M_i denote the market capitalization of asset i. The portfolio weights are then given by

$$w_i = \frac{M_i}{\sum_{j=1}^{N} M_j}. \tag{4.31}$$

This approach reflects the composition of the market portfolio and is widely used in practice, particularly in index construction (Hayes, 2025). Market capitalization weighting implicitly assigns larger weights to more established firms, which often exhibit lower volatility. However, this can lead to concentration in a small number of large companies and may reduce diversification from a risk perspective.

4.9.3 Risk parity portfolios

Risk parity weighting aims to allocate capital such that each asset contributes equally to the total portfolio risk. Let Σ denote the covariance matrix of asset returns and w the vector of portfolio weights. The total portfolio volatility is given by

$$\sigma_p = \sqrt{w^\top \Sigma w}. \tag{4.32}$$

The risk contribution of asset i is defined as

$$RC_i = \frac{w_i (\Sigma w)_i}{\sigma_p}. \tag{4.33}$$

A risk parity portfolio is constructed such that

$$RC_1 = RC_2 = \cdots = RC_N. \tag{4.34}$$

In practice, this implies assigning lower weights to more volatile assets and higher weights to less volatile ones. Compared to equal weighting, risk parity achieves a more balanced distribution of risk across assets, taking both volatility and correlations into account.

4.10 Statistical tools

The following section describes various statistical tools used in this study.

4.10.1 Statistical t-test

The t-test is a statistical method used to determine whether there exists a statistically significant difference between two sample means. Several variants of the t-test exist depending on the structure of the data and the relationship between the samples. Common forms include the one-sample t-test, the independent samples t-test, and the paired samples t-test.

In the context of comparing portfolio construction methods, the paired samples t-test is the most appropriate. This is because the returns of the portfolios are observed over the same time periods, meaning that each observation forms a natural pair. Since all portfolios are exposed to the same market conditions at a given time, their returns are not independent. Market-wide movements tend to affect all portfolios simultaneously, creating correlation between the observations.

A paired samples t-test is used to determine whether the mean difference between two related samples is statistically significantly different from zero. The test is appropriate when observations are naturally paired, such as measurements taken at the same time or under similar conditions. Instead of comparing two independent samples, the paired t-test is applied to the differences between the paired observations, effectively reducing the problem to a one-sample test on the difference series (Wooldridge, 2013).

4.10.1.1 Hypothesis testing

Hypothesis testing provides a statistical framework for evaluating whether observed differences in sample data reflect genuine underlying effects or are likely to have occurred by random variation. In the context of this study, hypothesis testing is used to assess whether differences in returns between portfolios constructed according to different rules are statistically significant (University of Sheffield, n.d.). The null hypothesis represents the absence of a systematic difference in performance, while the alternative hypothesis implies that one portfolio construction method yields a different average return than the other.

To conduct the test, the time series of return differences between two portfolios is defined as $d_t = R_t^A - R_t^B$. The hypothesis test then evaluates whether the expected value of these differences is equal to zero. If the null hypothesis is rejected, it indicates that the average return difference between the portfolios is statistically significant, suggesting that the portfolio construction methods lead to systematically different performance. The null hypothesis and the alternative hypothesis can be formulated as

$$H_0 : \mathbb{E}[d_t] = 0, \qquad H_1 : \mathbb{E}[d_t] \neq 0.$$

4.10.1.2 p-value

The p-value provides a quantitative measure for assessing the statistical significance of the test result. It represents the probability of observing a test statistic at least as extreme as the one obtained, under the assumption that the null hypothesis is true. (University of Sheffield, n.d.)

In the context of this study, the p-value is computed based on the t-statistic defined in Equation 4.38. A small p-value indicates that the observed average return difference is unlikely to have occurred due to random variation alone, providing evidence against the null

hypothesis.

Formally, for a two-sided test, the p-value is given by

$$p = 2 \cdot P\left(|T| \geqslant |t_{\text{obs}}|\right), \tag{4.35}$$

where t_{obs} denotes the observed t-statistic and T follows a t-distribution (or asymptotically a normal distribution when the sample size is large) (OpenStax, 2019).

A significance level α is chosen to determine the decision rule of the test. If $p < \alpha$, the null hypothesis is rejected, indicating a statistically significant difference in mean returns between the portfolios. Conversely, if $p \geqslant \alpha$, there is insufficient evidence to reject the null hypothesis.

4.10.1.3 Newey-West

To assess whether differences in portfolio performance are statistically significant, mean return differences between portfolio strategies are evaluated using Newey–West adjusted t-statistics. The Newey–West estimator provides heteroskedasticity and autocorrelation consistent standard errors, which are appropriate for financial return series that typically exhibit time-varying volatility and serial correlation. This allows for robust inference when testing whether the average return difference between two portfolio construction methods is significantly different from zero.

Heteroskedasticity occurs when the variance of the residuals varies across observations or over time. In financial time series, this phenomenon often appears as volatility clustering, where periods of high volatility tend to be followed by further periods of high volatility, while calm periods are followed by relatively low volatility. In such cases, the assumption of constant variance is violated. (Wooldridge, 2013)

Another common feature of time series data is autocorrelation (or serial correlation), which arises when residuals are correlated across time. This implies that the error term in one period may be related to error terms in previous periods. As a consequence, the assumption that residuals are independent is no longer valid. (Wooldridge, 2013)

When both heteroskedasticity and autocorrelation are present, the variance–covariance matrix of the error terms is no longer diagonal. Instead, it contains time-varying variances on the diagonal and non-zero covariances between observations across time. Under these circumstances, conventional standard error estimators are inconsistent. To obtain reliable statistical inference, heteroskedasticity and autocorrelation consistent (HAC) estimators are required, of which the Newey–West estimator is commonly used (Wooldridge, 2013).

When the theory in this section and section 4.10.1 is combined the t-test can be constructed as follows.

Let R_t^A and R_t^B be observed returns for two portfolios constructed with different methods.

$$d_t = R_t^A - R_t^B, \tag{4.36}$$

$$\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t. \tag{4.37}$$

To handle the presence of heteroskedasticity and auto-correlation the variance for $\bar{d}$ is calculated with Newey-West method. This gives the following formula for the t-statistic

$$t = \frac{\bar{d}}{\sqrt{Var_{NW}(\bar{d})}}. \tag{4.38}$$

To perform the test it is necessary to calculate the Newey-West standard error, which is the denominator in 4.38.

The Newey–West estimator adjusts the variance of the sample mean by incorporating weighted autocovariances of the return difference series. As described in Wooldridge (2013), the Newey–West variance estimator for $\bar{d}$ can be written as

$$Var_{NW}(\bar{d}) = \frac{1}{T}\left(\gamma_0 + 2\sum_{k=1}^{L} w_k \gamma_k\right). \tag{4.39}$$

where T denotes the number of observations, L is the chosen lag length, and w_k are lag-dependent weights defined as

$$w_k = 1 - \frac{k}{L+1}. \tag{4.40}$$

The term γ_k represents the sample autocovariance of the return difference series at lag k and is given by

$$\gamma_k = \frac{1}{T}\sum_{t=k+1}^{T}(d_t - \bar{d})(d_{t-k} - \bar{d}), \quad k = 1, 2, \ldots, L, \tag{4.41}$$

while γ_0 corresponds to the sample variance of the return difference series

$$\gamma_0 = \frac{1}{T}\sum_{t=1}^{T}(d_t - \bar{d})^2. \tag{4.42}$$

The Newey–West standard error is then obtained as

$$SE_{NW}(\bar{d}) = \sqrt{Var_{NW}(\bar{d})}, \tag{4.43}$$

which is used in Equation 4.38 to compute the t-statistic.

4.11 Monte carlo generation

Monte Carlo methods are computational techniques used to generate large numbers of observations for analytical purposes. Rather than relying on closed-form analytical solutions, these methods approximate the distribution of possible outcomes through repeated random sampling. By simulating many realizations of a random experiment, Monte Carlo methods allow researchers to estimate statistical properties such as expected values, variances, and higher moments (Glasserman, 2004). In finance, Monte Carlo techniques are widely applied to evaluate asset pricing models, assess portfolio risk, and analyze the statistical properties of investment strategies under uncertainty.

5 Method

This chapter outlines the method employed in the study. It describes the data collection process, including the selection of firms, sample period, and data sources, followed by the procedures used to clean, structure, and transform the data into analyzable formats. In addition, the chapter presents the construction of pooled datasets and the implementation of statistical measures used to characterize return distributions and wealth creation.

The chapter further details the portfolio construction process, including the design of rule-based and randomly generated portfolios, as well as the application of different weighting schemes. Finally, the evaluation framework is introduced, combining simulation-based benchmarking and statistical inference to assess whether systematic portfolio construction rules lead to statistically significant differences in average return.

5.1 Data collection

This section describes the data used in the empirical analysis and how it was collected.

5.1.1 Data source

Return data for publicly traded common stocks are collected from the Refinitiv Workspace database. Refinitiv is a widely used financial data provider that offers comprehensive and reliable market data. The data consists of time series of the TRI described in 4.3 and forms the empirical basis of the study and enables the examination of pooled return distributions and their statistical characteristics in Nordic equity markets. As the objective of this study is to examine portfolio returns in certain market environments, the analysis relies exclusively on quantitative market data and does not require qualitative information.

The datasets are initially exported from Refinitiv Workspace using the Datastream function and imported into Microsoft Excel for preliminary data preparation.

5.1.2 Sample selection

The empirical analysis is based on publicly traded companies listed on Nordic equity exchanges. The sample includes firms that were listed on at least one of the selected Nordic markets during the sample period. By including both currently listed and historically listed firms, the dataset captures the full population of traded securities and avoids survivorship bias that could arise from excluding firms that have been delisted.

The markets included in the study are Nasdaq Stockholm, Spotlight Stock Market, Nasdaq Copenhagen, Euronext Oslo, Nasdaq Helsinki, and Nordic Growth Market. Together, these exchanges represent the major trading venues for equities in the Nordic region. The market index used in the study is STOXX Nordic which represents large, mid, and small cap firms in the Nordic market (Sweden, Norway, Denmark, and Finland). A proxy for the risk-free rate, the 3-month EURIBOR, is also collected to enable the calculation of excess returns and wealth creation.

The sample is defined based on the exchange on which the security is traded rather than the country of incorporation of the firm. Consequently, some companies in the dataset may be headquartered outside the Nordic region while still being listed on a Nordic exchange. From an investment perspective, this distinction is appropriate, as investors typically access firms through the exchange on which their securities are traded. For each firm in the sample, time series data for the TRI are collected over the available listing history. This includes firms that were later delisted due to bankruptcy, acquisition, merger, or other corporate events. Including these firms ensures that the dataset reflects the full distribution of realized investment outcomes in the Nordic equity markets.

5.1.3 Sample period

The dataset used in this study covers multiple time horizons depending on the empirical analysis conducted. For the analysis of the empirical return distributions, one-year and five-year returns are calculated over the period from January 1, 1992 to January 1, 2026. This period is selected in order to provide a sufficiently large number of return observations while maintaining relevance to the modern structure of Nordic equity markets.

For the analysis of wealth creation, a longer historical period is considered. The wealth creation calculations begin in 1982, which corresponds to the earliest available observations in the Refinitiv Workspace database. Firm-level wealth creation is then tracked from the beginning of the available data until January 1, 2026. Using a longer time horizon for wealth creation allows the study to capture the cumulative economic contribution of firms over their entire time available in the database, while the shorter horizon used for the return distribution analysis is considered to be enough to get a reasonable amount of observations to draw conclusions regarding the pooled distributional properties of the annual equity returns.

5.2 Data processing

This section describes how the collected data is processed and prepared for analysis.

5.2.1 Cleaning and structuring the data

The raw data is first organized into separate Excel sheets according to country and time horizon. Annual TRI data is collected for each company and stored in the dedicated sheets. In addition, lifetime return data is retrieved and stored separately.

Daily data is retrieved and stored in separate sheets as well. This dataset includes both market index data (STOXX Nordic) and TRI data for individual stocks. The date series are carefully aligned across all datasets to ensure temporal consistency, which is required for the subsequent implementation of the framework described in Section 4.8, the implication of time in the skewness of return distribution.

5.2.2 Handling of delisted companies

The Refinitiv Workspace database does not provide a direct filter indicating the reason for a firm's delisting. This information is necessary for the study, as described in Section 3.3, and therefore the reason for delisting must be identified for each company that exited the market during the sample period. This is done by individually verifying each delisted firm and updating the dataset accordingly. By incorporating delisted firms and accounting for the reason for their removal from the market, the dataset avoids survivorship bias.

After the reason for delisting has been verified for each delisted company, the corresponding return series are also incorporated into the pooled datasets with the returns for the active companies.

5.2.3 Handling of missing observations

Financial datasets frequently contain missing observations due to inactive trading periods, incomplete historical coverage, or corporate events. Prior to the return calculations, the dataset is inspected for missing or invalid values. Observations with incomplete TRI data, mainly companies whose TRI was missing and resulted in errors, are removed from the analysis in order to avoid distortions in the return calculations. In cases where firms have incomplete time series due to listing or delisting events, returns are calculated only for the periods during which the firm was actively listed on the exchange.

5.2.4 Calculation of returns

Once the data has been cleaned and structured, the annual arithmetic returns for all companies in the Nordic market are calculated using equation 4.1 and pooled into a new dataset. This is first done for each market separately and later all markets are pooled together. Daily returns are calculated using the TRI observations for consecutive trading days. The daily return is therefore computed using the TRI values at time t and $t-1$ according to equation 4.1 where t denotes the trading day.

Based on the annual data, five-year returns are subsequently calculated using a rolling window approach. The same return equation is applied as for the annual returns, but the time interval between observations is extended from one year to five years. In practice, the five-year return is calculated using the TRI values at time t and $t+5$, capturing the cumulative return over the five-year holding period. Using a rolling window allows the five-year return series to include overlapping observations, meaning that returns are calculated for the periods t to $t+5$, $t+1$ to $t+6$, $t+2$ to $t+7$, and so forth. This procedure increases the number of available observations and enables a more comprehensive analysis of medium-term investment outcomes.

Lifetime returns are calculated using the full TRI series available for each firm. Since the TRI is normalized to start at 100 when a firm first appears in the dataset, the lifetime return is calculated as the ratio between the final TRI value and the initial index level. For firms that remain listed at the end of the sample period, the lifetime return is computed using

the TRI value observed on 01-01-2026 as the terminal value. The lifetime return for firm i is therefore calculated as

$$R_i^{lifetime} = \frac{TRI_{i,T}}{100} - 1, \tag{5.1}$$

where $TRI_{i,T}$ represents the final available TRI value for firm i.

5.2.5 Construction of pooled datasets

After returns have been calculated for each firm and market, the observations are combined into pooled datasets. First, pooled return datasets are constructed separately for each Nordic market in order to examine market-specific characteristics. In a second step, all markets are combined into a single pooled dataset representing the Nordic equity market as a whole. Each observation in the pooled dataset corresponds to the return of a specific firm over a given time horizon. This structure allows the empirical return distribution to be analyzed across both firms and time periods.

5.2.6 Construction of dataset used for portfolio construction

For the firms included in the pooled dataset, daily returns based on the TRI are extracted from Refinitiv Workspace together with annual market capitalization data. Market capitalization is used to classify firms into size-based segments. The portfolios are rebalanced on a yearly basis, which implies that the size-based segmentation must also be updated annually to ensure consistency in the set of firms included in each portfolio. At each rebalancing point, firms are sorted in ascending order based on their market capitalization and divided into three equally sized groups. The largest one third of firms are classified as large-cap, the middle one third as mid-cap, and the smallest one third as small-cap. This classification ensures a consistent segmentation of the market and allows for a comparative analysis of portfolio performance across different size categories.

5.3 Modelling

For the modelling section of this study, Python is used as the main tool for analyzing the data.

5.3.1 Visualization and statistical properties of the pooled return distribution

The pooled data sets are imported into Python, where the distribution of returns is visualized using histograms and QQ-plots. In addition to visual inspection, key descriptive statistics are calculated, including the mean, median, standard deviation, and skewness of the pooled return distributions. These measures provide a quantitative summary of the central tendency, dispersion, and asymmetry of returns and complement the graphical analysis. The analysis is conducted for one-year, five-year and lifetime return horizons in order to capture differences in the distribution of pooled returns across investment periods.

5.3.2 Calculation of wealth creation

In addition to analyzing the statistical distribution of stock returns, the study also examines how long-term value creation is distributed across firms in the Nordic equity markets. The concept of wealth creation follows the framework described in Section 4.7, where wealth creation measures the monetary value generated by a firm's equity in excess of a risk-free investment (EURIBOR 3M). The empirical implementation is based on firm-level market capitalization and return data obtained from Refinitiv Workspace. For each firm i and period t, wealth creation is calculated using Equation 4.10, which measures the value generated for investors relative to the risk-free rate. Market capitalization at the beginning of the period is

multiplied by the firm's excess return during that period.

To ensure comparability across time, each period's wealth creation is expressed in end-of-sample value by compounding with the risk-free rate as described in Equation 4.11. This adjustment allows wealth generated in different years to be measured in common monetary units. The total wealth creation of firm i is obtained by aggregating these adjusted values over the entire lifetime of the firm within the sample period. Firms that were delisted during the sample are included until their final trading period, ensuring that wealth creation reflects the full set of realized investment outcomes.

Finally, total market wealth creation is calculated by summing the cumulative wealth creation across all firms in the sample according to Equation 4.12. By comparing each firm's wealth contribution to the total market wealth creation, it becomes possible to evaluate the degree of concentration in long-term value creation and to identify the firms that account for the largest share of aggregate market wealth.

5.3.3 Time aspect of the skewness in return distributions

Daily arithmetic returns are computed in Python for each asset i and for the market index. For each asset, only the time period during which both asset and market return data are available is considered, ensuring consistency across observations. The expected returns are approximated using sample averages, as defined in equations 4.16 and 4.15.

Volatility is then estimated as the sample standard deviation of returns for both individual assets and the market, following equations 4.18 and 4.17. Based on these estimates, the covariance between each asset and the market is computed according to equation 4.19. The systematic risk of each asset is subsequently summarized by its beta coefficient, defined in equation 4.20.

Using the estimated beta values, the expected return for each asset is calculated within the Capital Asset Pricing Model (CAPM) framework, as specified in equation 4.21. These expected returns are then transformed into expected log returns using the approximation given in equation 4.22. Log returns are thereafter modeled according to equation 4.23.

Finally, total log returns are aggregated over the full observation period as described in Equation 4.24. Based on these aggregated values, the stochastic component of returns is estimated and normalized. This enables an analysis of the underlying sources of skewness in the return distribution.

5.3.4 Portfolio construction

The next step of the study focuses on portfolio construction. This section outlines how the portfolios are formed, considering both random and rule-based selection approaches. This enables a comparison between naive diversification and more informed, characteristic-based strategies. For this part of the study, data spanning the past ten years will be used.

The initial step in the portfolio construction process is to classify firms according to their market capitalization into small-, mid-, and large-cap segments. Within each segment, portfolios of varying sizes, 10, 50, and 100 stocks, are constructed based on different selection rules.

In addition to the selection of constituent stocks, the weighting scheme plays a central role in portfolio construction. In this study, portfolios are constructed using three different weight-

ing approaches: equal weighting, market capitalization weighting, and risk parity weighting.

Equal weighting assigns identical weights to all constituents, providing a simple benchmark and emphasizing diversification. Market capitalization weighting assigns weights proportional to firm size, thereby reflecting the structure of the underlying market. In contrast, the risk parity approach allocates weights such that each asset contributes equally to the overall portfolio risk, allowing for a more balanced distribution of risk across holdings.

Following the principles outlined in this section, the portfolios are rebalanced annually to maintain consistency in the applied construction rules over time.

5.3.4.1 Skew strategy

One of the rule-based approaches considered is a skewness-based strategy. To examine how skewness in individual stock returns affects portfolio performance, portfolios are constructed by selecting firms exhibiting the highest levels of return skewness as one rule, and firms exhibiting the lowest levels of return skewness as another.

Within each market capitalization segment, firms are ranked according to their historical skewness. Stocks are sorted in descending order based on the estimated skewness measure, after which portfolios are formed according to the baseline structure described in 5.3.4.

5.3.4.2 Momentum strategy

In addition to the skewness-based approach, a momentum-based strategy is implemented to examine how past return performance influences portfolio outcomes. Within each market capitalization segment, firms are ranked according to their historical returns over the previous year. Stocks are sorted in descending order based on these returns, and portfolios are constructed by selecting the top-performing firms. The portfolio formation follows the same baseline structure as previously described, with portfolios consisting of 10, 50, and 100 stocks.

This approach allows for a direct comparison between portfolios formed on past performance and those based on distributional characteristics such as skewness, while maintaining consistency in portfolio size and segmentation.

5.3.4.3 Portfolio implementation

Portfolio construction is carried out in a two-step procedure consisting of stock selection and portfolio weighting. This separation allows the analysis to distinguish between the effects of selection rules and weighting schemes. At each annual rebalancing date, stocks are first selected based on predefined selection rules. Specifically, stocks are ranked according to their historical skewness and momentum, calculated using past return data. Based on these rankings, portfolios are formed by selecting stocks with the highest skewness, lowest skewness, or highest momentum, depending on the portfolio specification.

Given the selected set of stocks, portfolio weights are then assigned according to one of three weighting schemes: equally weighted, market capitalization weighted, or volatility-based (risk parity) weighting.

For equally weighted portfolios, each stock is assigned an equal share of the portfolio according to

$$w_{i,t} = \frac{1}{N},$$

where N is the number of stocks in the portfolio.

For market capitalization weighted portfolios, weights are proportional to firm size according to

$$w_{i,t} = \frac{M_{i,t}}{\sum_{j=1}^{N} M_{j,t}},$$

where $M_{i,t}$ denotes the market capitalization of stock i at time t.

For risk parity portfolios, weights are determined such that each asset contributes equally to the total portfolio risk. Let Σ_t denote the covariance matrix of asset returns at time t, and w_t the vector of portfolio weights. The portfolio volatility is given by

$$\sigma_{p,t} = \sqrt{w_t^\top \Sigma_t w_t}.$$

The risk contribution of asset i is defined as

$$RC_{i,t} = \frac{w_{i,t}(\Sigma_t w_t)_i}{\sigma_{p,t}}.$$

The risk parity portfolio is constructed such that

$$RC_{1,t} = RC_{2,t} = \cdots = RC_{N,t}.$$

At the beginning of the first period, the portfolio value is normalized to one. The allocated weights determine the initial capital invested in each stock. During each period, the portfolio evolves according to the returns of its constituent stocks. The portfolio return in period t is calculated as the weighted sum of individual stock returns

$$R_{p,t} = \sum_{i=1}^{N} w_{i,t} R_{i,t},$$

where $R_{i,t}$ denotes the return of stock i during period t.

The total portfolio value is updated by compounding returns over time as

$$V_{p,t} = V_{p,t-1}(1 + R_{p,t}).$$

At the end of each year, the portfolio is rebalanced. This involves re-estimating skewness and momentum measures using updated historical data, selecting a new set of stocks, and reallocating the total portfolio value according to the chosen weighting scheme. The updated portfolio value is then equally or proportionally redistributed among the selected stocks, depending on the weighting method.

This process is repeated annually over the 10-year evaluation period, resulting in a time series of portfolio returns for each portfolio specification.

5.3.5 Monte carlo simulation of random portfolios

To construct random portfolios that are directly comparable to the rule-based portfolios, a Monte Carlo simulation approach is employed. The simulated portfolios are generated under the same structural constraints as the rule-based portfolios, ensuring consistency across all relevant dimensions. Specifically, portfolios are formed with varying levels of diversification, consisting of 10, 50, and 100 constituents, drawn from the small-, mid-, and large-cap segments.

For each combination of size segment, diversification level, and weighting scheme, 10 000 random portfolios are generated. This results in a comprehensive benchmark distribution for each portfolio specification. The simulated portfolios are thus identical to the rule-based portfolios in all respects such as investment universe, number of constituents, and weighting scheme except for the selection mechanism, which is random rather than based on a predefined rule. The simulation will be performed using Python scripts on the data fetched from Excel.

5.4 Evaluation

This section describes how the constructed rule-based portfolios are evaluated against different benchmarks and with each other.

5.4.1 Benchmarking the rule-based portfolios against the randomly constructed portfolios

To evaluate whether the performance of the rule-based portfolios can be attributed to systematic selection rules rather than random chance, each portfolio is benchmarked against a distribution of randomly constructed portfolios. The random portfolios are generated using the Monte Carlo simulation procedure described in Section 5.3.5. This simulation provides an empirical distribution of portfolio returns that reflects the range of outcomes achievable through random selection under identical constraints.

The return of each rule-based portfolio is evaluated relative to this distribution. In particular, the percentile rank of the rule-based portfolio is calculated, as well as the proportion of randomly constructed portfolios that achieve higher returns over the evaluation period. This approach allows for an assessment of whether the observed performance of the rule-based portfolios lies within the range of outcomes expected by chance or represents a statistically and economically meaningful deviation from random portfolio construction.

In addition to percentile-based evaluation, the performance of each rule-based portfolio is also compared to the mean and median returns of the randomly constructed portfolios. While the mean reflects the average outcome of the simulated portfolios, the median provides a measure of the typical outcome and is less sensitive to extreme observations. This distinction is particularly relevant in the presence of skewed return distributions, where a small number of extreme outcomes may have a substantial impact on the mean. Comparing rule-based portfolios to both measures therefore provides a more comprehensive assessment of their relative performance.

5.4.2 Benchmarking the rule-based portfolios against the market index and the risk free rate

To further provide an economically meaningful evaluation of the rule-based portfolios, their performance is benchmarked against both a broad market index and the risk-free rate. This allows for an assessment of whether the portfolio construction methods generate excess returns relative to passive investment alternatives and a baseline return with no risk exposure.

The STOXX Nordic index serves as a natural benchmark, representing the aggregate performance of the investable universe. Comparing the rule-based portfolios to the market index enables an evaluation of whether the selection rules add value beyond a passive buy-and-hold strategy. The comparison is conducted by evaluating the total return of the index alongside the returns of the rule-based portfolios over the same time period, ensuring

consistency in the performance assessment.

In addition, the risk-free rate, 3-month EURIBOR, is used as a reference point to assess whether the portfolios deliver compensation for bearing risk. By comparing portfolio returns to the risk-free rate, it is possible to evaluate whether the strategies generate positive excess returns. For the comparison, the total return obtained by continuously reinvesting at the risk-free rate is calculated and evaluated against the total return of the rule-based portfolios over the same time period.

This comparison ensures that the results are interpretable both from a statistical and an economic perspective, allowing for a comprehensive assessment of the effectiveness of the portfolio construction methods.

5.4.3 Statistical test

To perform t-tests in accordance with the framework described in Section 4.10.1, portfolios are compared in a controlled manner where only one dimension is allowed to vary at a time. Rule-based portfolios are evaluated through a series of pairwise comparisons, where one portfolio characteristic is varied at a time while all others are held constant. Specifically, three types of comparisons are conducted.

First, portfolios constructed under the same selection rule, weighting scheme and have the same degree of diversification are compared across different market capitalization segments, allowing for an analysis of size-related effects.

Second, portfolios constructed under the same selection rule and within the same market capitalization segment are compared across different weighting schemes, isolating the impact of capital allocation.

Third, portfolios with identical market capitalization segments and weighting schemes are compared across different selection rules, enabling an evaluation of the effect of the portfolio construction rule itself.

The analysis is based on daily return data, ensuring a consistent and high-frequency comparison across portfolios. For each pair of portfolios, the time series of return differences is computed using their daily returns. Since both portfolios are observed over the same time periods, they are exposed to identical market conditions, making the comparison robust.

5.4.3.1 Return differences

To evaluate differences in performance between portfolios, pairwise comparisons are conducted based on their daily returns. For each pair of portfolios, the time series of return differences is constructed. Let $R_{p,t}^{(A)}$ and $R_{p,t}^{(B)}$ denote the daily returns of portfolio A and portfolio B at time t, respectively. The return difference is then defined as

$$d_t = R_{p,t}^{(A)} - R_{p,t}^{(B)}.$$

This results in a time series $\{d_t\}_{t=1}^{T}$ representing the daily excess return of one portfolio relative to another. Since both portfolios are observed over the same time periods, the difference d_t reflects performance differences under identical market conditions. The mean of this series,

$$\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t,$$

represents the average daily return difference between the two portfolios and serves as the basis for subsequent statistical testing.

5.4.3.2 Newey-West adjusted t-statistic

To assess whether the average return difference is statistically significant, a t-test is performed on the time series of return differences. However, since daily financial returns may exhibit autocorrelation and heteroskedasticity, standard t-statistics may lead to biased inference. To address this, Newey-West adjusted standard errors are employed.

Let d_t denote the return difference defined in the previous Subsection 5.4.3.1, and let $\bar{d}$ denote its sample mean. The t-statistic is given by

$$t = \frac{\bar{d}}{\sqrt{Var_{NW}(\bar{d})}},$$

where $Var_{NW}(\bar{d})$ is the Newey-West estimator of the variance of the mean. The Newey-West variance estimator is defined as

$$Var_{NW}(\bar{d}) = \frac{1}{T}\left(\gamma_0 + 2\sum_{k=1}^{L}\left(1 - \frac{k}{L+1}\right)\gamma_k\right),$$

where T is the number of observations, k denotes the lag index, and L is the chosen lag length. In this study, the lag length L is selected based on a standard rule-of-thumb depending on the sample size, ensuring an appropriate balance between bias and variance (Wooldridge, 2013). L is defined as

$$L = \lfloor 4(T/100)^{2/9} \rfloor.$$

The term γ_k represents the sample autocovariance at lag k, defined as

$$\gamma_k = \frac{1}{T}\sum_{t=k+1}^{T}(d_t - \bar{d})(d_{t-k} - \bar{d}).$$

The lag length L determines the number of autocovariances included in the estimator and controls the degree of adjustment for serial correlation. The weights $\left(1 - \frac{k}{L+1}\right)$ ensure that higher-order lags receive less influence. By incorporating these adjustments, the Newey-West estimator provides consistent standard errors in the presence of both heteroskedasticity and autocorrelation, allowing for reliable statistical inference. This procedure is done for each compared pair of portfolios.

5.4.3.3 Hypothesis testing and p-values

To evaluate whether the observed return differences are statistically significant, hypothesis testing is conducted based on the t-statistics derived in the previous subsection.

The null hypothesis is defined as

$$H_0 : \mathbb{E}[d_t] = 0,$$

implying that there is no significant difference in average returns between the two portfolios. The alternative hypothesis is given by

$$H_1 : \mathbb{E}[d_t] \neq 0,$$

The statistical significance of the estimated mean return difference is evaluated using the t-statistic obtained from the Newey-West adjusted standard errors. The p-value is computed as

$$p = 2 \cdot P\left(|T| \geqslant |t_{\text{obs}}|\right),$$

The p-value represents the probability of observing a test statistic at least as extreme as the one obtained, under the null hypothesis. If the p-value is below a predefined significance level (e.g., 5%), the null hypothesis is rejected, indicating a statistically significant difference in portfolio performance. If the p-value exceeds the significance level, the null hypothesis cannot be rejected. This framework allows for a systematic evaluation of whether differences in average portfolio returns are driven by the applied selection rules, weighting schemes, or market segmentation, rather than random variation.

The test evaluates whether there is a statistically significant difference in the average returns between the two portfolios. The sign of the estimated mean return difference determines the direction of the performance comparison. A positive and statistically significant mean return difference indicates that portfolio A achieves higher average returns than portfolio B. Conversely, a negative and statistically significant mean return difference implies that portfolio B achieves higher average returns than portfolio A.

6 Results

This chapter presents the results. Note that the results are presented factually, striving for objectivity as far as possible. The results shall not be analyzed, discussed or evaluated. This is left for the discussion chapter.

In case the method chapter has been divided into subheadings such as pre-study, implementation and evaluation, the result chapter should have the same sub-headings. This gives a clear structure and makes the chapter easier to write.

In case results are presented from a process (e.g. an implementation process), the main decisions made during the process must be clearly presented and justified. Normally, alternative attempts, etc, have already been described in the theory chapter, making it possible to refer to it as part of the justification.

7 Discussion

This chapter contains the following sub-headings.

7.1 Results

Are there anything in the results that stand out and need be analyzed and commented on? How do the results relate to the material covered in the theory chapter? What does the theory imply about the meaning of the results? For example, what does it mean that a certain system got a certain numeric value in a usability evaluation; how good or bad is it? Is there something in the results that is unexpected based on the literature review, or is everything as one would theoretically expect?

7.2 Method

This is where the applied method is discussed and criticized. Taking a self-critical stance to the method used is an important part of the scientific approach.

A study is rarely perfect. There are almost always things one could have done differently if the study could be repeated or with extra resources. Go through the most important limitations with your method and discuss potential consequences for the results. Connect back to the method theory presented in the theory chapter. Refer explicitly to relevant sources.

The discussion shall also demonstrate an awareness of methodological concepts such as replicability, reliability, and validity. The concept of replicability has already been discussed in the Method chapter (5). Reliability is a term for whether one can expect to get the same results if a study is repeated with the same method. A study with a high degree of reliability has a large probability of leading to similar results if repeated. The concept of validity is, somewhat simplified, concerned with whether a performed measurement actually measures what one thinks is being measured. A study with a high degree of validity thus has a high level of credibility. A discussion of these concepts must be transferred to the actual context of the study.

The method discussion shall also contain a paragraph of source criticism. This is where the authors' point of view on the use and selection of sources is described.

In certain contexts it may be the case that the most relevant information for the study is not to be found in scientific literature but rather with individual software developers and open

source projects. It must then be clearly stated that efforts have been made to gain access to this information, e.g. by direct communication with developers and/or through discussion forums, etc. Efforts must also be made to indicate the lack of relevant research literature. The precise manner of such investigations must be clearly specified in a method section. The paragraph on source criticism must critically discuss these approaches.

Usually however, there are always relevant related research. If not about the actual research questions, there is certainly important information about the domain under study.

7.3 The work in a wider context

There must be a section discussing ethical and societal aspects related to the work. This is important for the authors to demonstrate a professional maturity and also for achieving the education goals. If the work, for some reason, completely lacks a connection to ethical or societal aspects this must be explicitly stated and justified in the section Delimitations in the introduction chapter.

In the discussion chapter, one must explicitly refer to sources relevant to the discussion.

8 Conclusion

This chapter contains a summarization of the purpose and the research questions. To what extent has the aim been achieved, and what are the answers to the research questions?

The consequences for the target audience (and possibly for researchers and practitioners) must also be described. There should be a section on future work where ideas for continued work are described. If the conclusion chapter contains such a section, the ideas described therein must be concrete and well thought through.

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